



KOGNIX GenAI Engine for the Enterprise Workspace

**A powerful way of working with
Generative AI**

KOGNIX GenAI Engine

Executive summary

In an era characterized by data proliferation and accelerated digital transformation, conventional productivity tools are increasingly inadequate. Organizations require intelligence systems that do more than automate tasks—they must deliver accuracy, accountability, and trust at scale.

The KOGNIX AI Engine represents the evolution from experimental AI deployments to a mission-critical **Grounded Intelligence Synthesis** (GIS) framework. Purpose-built to overcome the long-standing “Garbage In, Garbage Out” (GIGO) limitations of earlier systems, KOGNIX enforces semantic integrity through Deep Data Understanding (DDU) and **Verified Knowledge Retrieval** (VKR), ensuring that outputs are grounded in authoritative, high-fidelity data.

At its core, the platform integrates an AI-native database optimized for high-dimensional workloads, enabling precise retrieval and contextual synthesis of enterprise knowledge. This architecture systematically eliminates hallucination risks that have historically restricted the use of Large Language Models in regulated and high-assurance environments.

Every response generated by KOGNIX AI Engine is anchored by verifiable audit anchors, creating a transparent and traceable context layer. This ensures full explainability and 100% auditability—critical requirements for enterprise decision-making, compliance, and governance.

Engineered for organizations where precision is non-negotiable, KOGNIX AI Engine embeds verified intelligence directly into daily operations. The result is faster execution, smarter decisions, reduced operational risk, and maximized return on investment—transforming AI from a productivity enhancer into a trusted enterprise capability.

KOGNIX AI PRINCIPLES

Deliver Responsible and Societal Value

Prevent Bias and Promote Fair Outcomes

Prioritize Safety, Reliability, and Robustness

Ensure Human Accountability and Oversight

Embed Privacy and Data Protection by Design

Uphold Scientific Rigor and Technical Excellence

Enable Ethical and Appropriate Use

this guide highlights existing and soon-to-be released features to help you plan and implement your generative ai journey with KOGNIX.

Redefining Productivity with KOGNIX AI Engine

Productivity in the AI era is no longer defined by working harder or automating isolated tasks. It is defined by intelligent collaboration between humans and machines—where AI operates with deep awareness of business context and institutional knowledge. The KOGNIX AI Engine redefines productivity by ensuring that every AI interaction is grounded in real, enterprise-grade knowledge rather than generic training data.

With KOGNIX AI Engine, employees no longer lose time navigating fragmented documents, disconnected systems, or organizational silos. Instead, they engage with a unified intelligence layer that understands organizational context, retrieves the most relevant information, and delivers actionable insights in real time. This approach enables:

Beyond incremental efficiency gains, KOGNIX fundamentally transforms knowledge work by automating its most labour-intensive components—document synthesis, cross-referencing, and structured data extraction. Organizations adopting advanced Grounded Intelligence Synthesis (GIS) architectures have reported 30–42% improvements in overall productivity and up to a 95% reduction in information retrieval time.

Through a re-engineered agent architecture, KOGNIX further optimizes routine business processes, reducing processing time by approximately 50% compared to earlier AI implementations. As a result, senior professionals can redirect up to 60% of their time away from administrative document management toward high-value strategic analysis, planning, and decision-making.

By converting enterprise knowledge into an always available, verified intelligence asset, KOGNIX AI Engine elevates productivity from task execution to sustained strategic impact.

How it Works: Architecture of Truth

The KOGNIX AI Engine operates through a sophisticated vision-based ingestion and retrieval pipeline:

- **Deep Data Understanding (DDU):** Unlike standard tools that strip formatting, KOGNIX utilizes vision models to perform Document Layout Recognition (DLR) and Table Structure Recognition (TSR). This ensures that complex tables, multi-column reports, and mixed-media documents retain their semantic integrity during ingestion.
- **Template-Based Chunking:** Data is not arbitrarily split; it is processed via specialized templates (e.g., Laws, Manuals, Financial Tables) that respect the original document's logic.
- **KOGNIX AI Native DB:** The core engine provides a "three-way hybrid search" (dense vector, sparse vector, and full-text) integrated with ColBERT-based tensor reranking. This ensures that technical terminology and specific data points are retrieved with sub-millisecond latency.
- **Verified Knowledge Retrieval (VKR):** The synthesis phase generates responses exclusively from retrieved data, providing Audit Anchors that allow users to verify the source of every claim with a single click.

What is a good prompt?

Use natural language: Write as if you're speaking to another person and express complete thoughts in full sentences.

Include important details: The best prompts include a persona, task, context, and format

Be concise: State your request in brief – but specific – language and avoid jargon

Make it a conversation: If the results don't meet your expectations, use follow-up prompts to fine-tune them.

How KOGNIX AI Engine Can Help Teams and Organizations

Boosting productivity with KOGNIX

Advancements in generative AI will trigger technological and cultural transformations comparable to previous phase-shift waves like personal computers, the internet, cloud computing, and mobile devices. As with any of these major technological shifts, generative AI will both reshape industry structures and create new markets and revenue opportunities.

Generative AI can create value for companies in several ways. Two of the most important ways are:

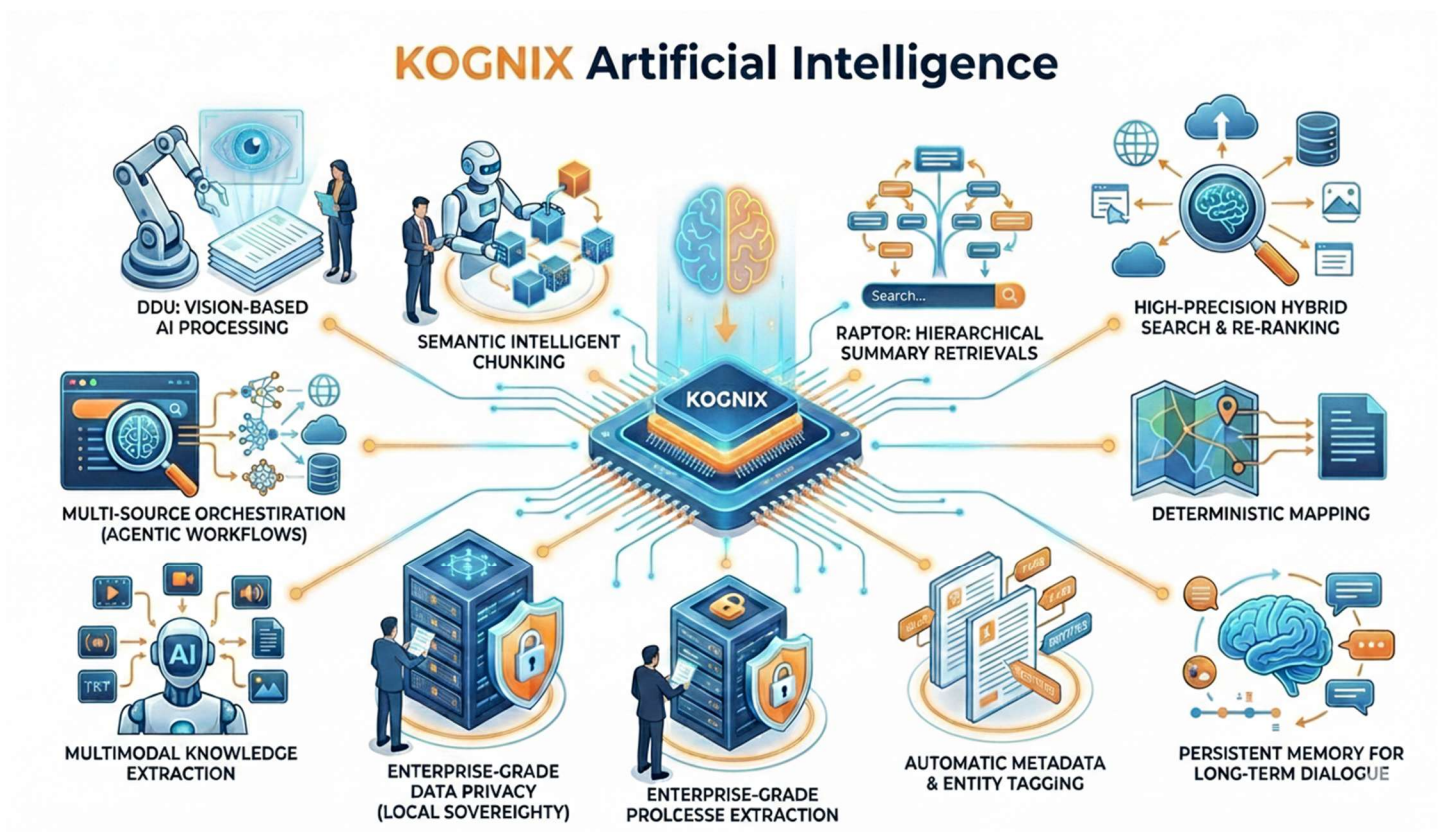
- **Focus employees on high-value activities**
Automating business processes and assisting with time-consuming or sometimes difficult tasks will make teams and individuals more productive.
- **Improving business outcomes**
Increasing employee productivity will boost the company's performance and can result in more revenue as more output is available to be commercialized.

KOGNIX functions as a centralized "intelligent core," breaking down data silos by gathering disparate information—from AWS S3 buckets to Notion wikis—into a unified knowledge index.

- **Knowledge Continuity:** It captures institutional memory, ensuring that critical data is not lost during personnel transitions.
- **Scalable Intelligence:** The system supports high-concurrency online scenarios, allowing teams to scale their AI-assisted workflows without performance degradation.

10 Capabilities to work Faster, Smarter & Accurately

1. DDU: Vision-Based AI Processing
2. Semantic Intelligent Chunking
3. RAPTOR: Hierarchical Tree Retrievals
4. High-Precision Hybrid Search & Re-ranking
5. Multi-Source Orchestration (Agentic Workflows)
6. Deterministic Mapping
7. Multimodal Knowledge Extraction
8. Enterprise-Grade Data Privacy (Local Sovereignty)
9. Automatic Metadata & Entity Tagging
10. Persistent Memory for Long-Term Dialogue



Deep Data Understanding

Most retrieval AI systems use simple text scrapers that ignore layout. KOGNIX AI Engine employs Grounded Intelligence Synthesis (GIS) framework, a sophisticated vision-based AI. It treats a PDF or document like an image first, using computer vision models to identify "zones" such as headers, footers, captions, and—crucially—complex table structures. It recreates the document's hierarchy in a digital format before the LLM ever sees it

This approach is significantly higher accuracy and reliability of retrieved information.

- **Preservation of Context and Meaning**
By retaining document layout and hierarchy, the system understands how content is organized. Headings, sections, footnotes, and captions are interpreted in context, reducing misinterpretation that often occurs when text is flattened.
- **Accurate Table and Data Extraction**
Complex tables are correctly reconstructed with proper row–column relationships. This prevents data corruption, misplaced values, or incorrect associations—a common failure point of traditional text scrapers.
- **Improved Retrieval Precision**
Structured segmentation enables more precise chunking and indexing. Queries can be matched against the most relevant sections rather than arbitrary text fragments, improving retrieval quality.
- **Reduced Hallucinations by the LLM**
When the LLM receives well-structured, semantically grounded input, it is less likely to infer or fabricate missing relationships, leading to more trustworthy outputs.
- **Stronger Traceability and Explainability**
Responses can be traced back to specific document zones (e.g., a table cell or subsection), supporting citations, audits, and enterprise compliance requirements.
- **Better Performance on Enterprise Data**
This approach is especially valuable for technical manuals, financial reports, research papers, and regulatory documents where layout carries meaning.

In summary, treating documents as visual structures rather than raw text ensures that the AI reasons over information as humans intended it to be read, resulting in more accurate, explainable, and enterprise-grade retrieval and generation.

Semantic Intelligent Chunking

Standard AI pipelines typically segment text into fixed-length chunks (for example, every 500 characters), often breaking sentences and fragmenting meaning. KOGNIX AI Engine instead applies semantic templates—such as General, Legal, Research Paper, Technical Manual etc, each with domain-specific chunking logic. It understands structural and semantic cues within the content. For instance, the Legal template identifies articles, sections, and clauses, ensuring that a legal rule and its associated exceptions remain within the same data block, preserving interpretive coherence.

Domain-aware chunking is substantial and directly impact the quality, reliability, and usability:

- **Preserved Semantic Integrity**
Keeping complete concepts—such as a legal rule and its exceptions—within a single chunk prevents loss of meaning that occurs when sentences or clauses are arbitrarily split.
- **Higher Retrieval Accuracy**
Queries are matched against logically complete units of information rather than partial fragments, resulting in more relevant and precise retrieval.
- **Improved Reasoning by the LLM**
When the LLM receives coherent, self-contained chunks, it can reason more effectively, reducing incorrect inferences and incomplete answers.
- **Reduced Hallucinations and Contradictions**
By avoiding fragmented context, the system minimizes situations where the LLM fills in gaps or overlooks critical qualifiers, leading to more trustworthy outputs.
- **Domain-Specific Performance Gains**
Templates tailored for legal, academic, or technical content respect domain conventions (e.g., clauses, citations, sections), which is essential for accuracy in regulated or complex fields.
- **Better Explainability and Compliance**
Responses can be traced back to meaningful document units (articles, sections, paragraphs), supporting citation, validation, and audit requirements.

In essence, semantic template-based chunking allows the system to retrieve and generate answers based on complete ideas rather than arbitrary text slices, delivering enterprise-grade accuracy and reliability.

RAPTOR: Hierarchical Tree Retrieval

RAPTOR (Recursive Abstractive Processing for Tree-Organized Retrieval) is a technique used by KOGNIX AI Engine to recursively summarize related content chunks and organize them into a hierarchical information tree. When a user submits a broad or high-level query, the system retrieves information from higher-level summary nodes within the tree, rather than relying solely on narrowly scoped, low-level text snippets. This enables more coherent and comprehensive responses to abstract or exploratory questions.

The use of RAPTOR-style hierarchical retrieval delivers several important benefits over flat, snippet-based approaches:

- **Stronger Performance on Broad and Abstract Queries**
High-level questions (e.g., “Summarize the compliance approach” or “Explain the system architecture”) are better answered using synthesized summaries rather than isolated granular chunks.
- **Improved Coherence and Narrative Flow**
Retrieving from summary nodes produces responses that are structured and logically complete, avoiding disjointed answers assembled from unrelated snippets.
- **Reduced Context Overload**
Instead of injecting many small chunks into the prompt, the system selects higher-level abstractions, optimizing token usage while preserving meaning.
- **Faster and More Efficient Retrieval**
Searching over a hierarchical tree narrows the retrieval space, reducing latency and improving scalability for large document collections.
- **Better Multi-Granularity Reasoning**
The model can operate at different levels of detail—starting with summaries and drilling down only when necessary—mirroring how humans explore complex information.
- **Lower Risk of Hallucination**
Abstracted, synthesized nodes provide clearer conceptual grounding, reducing the likelihood that the LLM invents connections between unrelated fragments.

In summary, tree-organized retrieval enables the system to answer both high-level and detailed questions more accurately, efficiently, and coherently than flat retrieval methods.

High-Precision Hybrid Search & Re-ranking

KOGNIX AI Engine employs a multi-path retrieval strategy that operates in parallel. It performs keyword-based search using BM25 to capture exact terminology, while simultaneously executing vector-based dense retrieval to identify semantically related content. The combined results are then processed by a cross-encoder re-ranker, which applies high-compute relevance scoring to precisely assess the relationship between the user query and each candidate document snippet.

Multi-path retrieval strategy with cross-encoder re-ranking is both practical and material to system performance:

- **Comprehensive Recall**
Combining keyword (BM25) and vector (dense) retrieval ensures that neither exact matches nor semantically relevant content are missed. This is especially important for technical terms, acronyms, and proper nouns.
- **Higher Precision Through Re-Ranking**
The cross-encoder evaluates the query and candidate passages jointly, enabling fine-grained relevance assessment that significantly improves the quality of the final retrieved results.
- **Reduced False Positives**
Dense retrieval alone can surface conceptually similar but contextually incorrect passages. The re-ranker filters these out by deeply analysing query–document alignment.
- **Improved Answer Accuracy**
Feeding the LLM with highly relevant, well-ranked context leads to more accurate, consistent, and grounded responses.
- **Robustness Across Query Types**
The system performs well for both narrow, exact queries (e.g., identifiers, codes) and broad, conceptual questions, without requiring the user to adjust query phrasing.
- **Enterprise-Grade Reliability**
This layered retrieval approach provides predictable and auditable relevance behaviour, which is critical for production, regulated, and mission-critical use cases.

In summary, multi-path retrieval with cross-encoder re-ranking balances recall and precision, delivering consistently high-quality context to the LLM and materially improving downstream generation outcomes.

Multi-Source Orchestration (Agentic Workflows)

KOGNIX AI Engine includes a Directed Acyclic Graph (DAG) editor that enables users to visually compose advanced workflows. Through a drag-and-drop interface, components such as Web Search, Python Executor, and Multi-LLM Evaluator can be connected to build sophisticated reasoning pipelines that extend well beyond basic question-and-answer interactions.

The use of a DAG-based workflow editor provides several strategic and operational benefits:

- **Composable and Modular Reasoning**
Complex tasks can be broken into discrete, reusable components, making reasoning pipelines easier to design, understand, and maintain.
- **Support for Advanced Use Cases**
By chaining tools such as web search, code execution, and multi-model evaluation, the system can handle analysis, validation, and decision-making scenarios that exceed the capabilities of simple Q&A flows.
- **Greater Transparency and Explainability**
Each step in the reasoning chain is explicit and inspectable, enabling teams to understand how an output was produced and to audit or debug individual stages.
- **Faster Iteration and Experimentation**
Drag-and-drop composition reduces development overhead, allowing non-specialist users to rapidly prototype, test, and refine workflows without extensive coding.
- **Improved Reliability and Control**
Deterministic execution paths reduce unpredictable behaviour, while fine-grained control over each node improves system robustness in production environments.
- **Scalability and Reuse**
Proven workflows can be standardized, versioned, and reused across teams or applications, accelerating adoption and ensuring consistency.

In effect, a DAG-based editor transforms the system from a single-step responder into a flexible reasoning platform capable of supporting complex, enterprise-grade AI workflows.

Deterministic Mapping

KOGNIX AI Engine enforces a precise metadata linkage between generated content and the original source locations, including exact X–Y coordinates within the source PDF. During response generation, the system cross-references LLM output tokens with the retrieved document chunks, enabling each statement to be traced back to a verifiable source location.

Maintaining strict metadata mapping between generated text and original source coordinates delivers several critical benefits:

- **End-to-End Traceability**
Every generated statement can be traced back to a precise location in the source document, supporting verification at the sentence, paragraph, or even table-cell level.
- **Higher Trust and Confidence**
Users can directly validate answers against the original content, increasing confidence in the system's outputs—particularly for high-stakes or regulated use cases.
- **Reduced Hallucination Risk**
By grounding generated tokens in retrieved chunks, the system discourages unsupported generation and minimizes the likelihood of fabricated or inferred information.
- **Auditability and Compliance Readiness**
Exact source coordinates enable rigorous audits, citations, and regulatory reviews, which are essential in legal, financial, healthcare, and enterprise environments.
- **Improved Explainability**
The system can explain not only *what* the answer is, but *where* it came from, aligning AI behaviour with enterprise transparency requirements.
- **Faster Error Detection and Correction**
When inaccuracies occur, teams can immediately pinpoint the originating source location, accelerating debugging and content remediation.

In summary, precise source-to-output mapping transforms KOGNIX AI Engine from a black-box generator into a verifiable, auditable, and enterprise-grade knowledge system.

Multimodal Knowledge Extraction

KOGNIX AI Engine extends beyond text by incorporating Vision–Language Models (VLMs). These models enable the system to interpret diagrams, flowcharts, and even handwritten notes, translating visual relationships into structured textual representations. The resulting descriptions are indexed and searchable in the same manner as conventional text, making visual content fully accessible to retrieval workflows.

Integrating Vision–Language Models delivers several meaningful advantages for retrieval and reasoning systems:

- **Access to Non-Textual Knowledge**
Critical information embedded in diagrams, flowcharts, schematics, and handwritten notes becomes machine-readable and searchable, eliminating blind spots in document understanding.
- **Improved Retrieval Coverage**
By converting visual relationships into structured text, the system ensures that visual content is retrievable alongside traditional paragraphs, increasing overall recall.
- **Better Understanding of Complex Concepts**
Many technical and operational ideas are conveyed more clearly through visuals. Interpreting these elements enables more accurate reasoning over processes, architectures, and workflows.
- **Unified Search and Indexing Experience**
Users can query across text and visuals using the same natural language interface, without needing to know where or how the information is represented.
- **Reduced Information Loss**
Traditional OCR often extracts labels but misses relationships. VLMs capture both entities and their connections, preserving the intent of the original visual.
- **Stronger Enterprise Use Cases**
This capability is particularly valuable for engineering diagrams, system architectures, research figures, compliance flowcharts, and handwritten field notes.

In effect, VLM integration ensures that visually expressed knowledge is treated as first-class information, enabling more complete, accurate, and actionable retrieval and generation.

Enterprise-Grade Data Privacy (Local Sovereignty)

KOGNIX AI Engine is designed to be infrastructure-agnostic. It can be deployed using Docker on private, on-premises servers and integrated with local LLM inference engines, ensuring that all data processing remains within the local network and never leaves the organization's environment.

Designing KOGNIX to be infrastructure-agnostic and fully operable on local networks provides several key benefits:

- **Data Privacy and Security**
Sensitive or proprietary information stays entirely on premises, reducing exposure to external networks and cloud providers.
- **Compliance with Regulations**
Organizations handling regulated data (e.g., healthcare, finance, government) can meet strict legal and industry compliance requirements, such as GDPR or HIPAA.
- **Reduced Latency**
Local deployment enables faster data retrieval and LLM inference since there is no dependency on external cloud services or internet connectivity.
- **Control and Customization**
Teams can configure hardware, inference engines, and system parameters to meet specific performance, cost, or security needs.
- **Operational Reliability**
Local deployments are less dependent on external service availability, minimizing downtime caused by network outages or cloud service disruptions.
- **Enterprise-Grade Flexibility**
Being infrastructure-agnostic allows integration with existing IT ecosystems, hybrid setups, or private cloud environments without locking into a single provider.

In short, local, infrastructure-agnostic deployment ensures secure, compliant, fast, and fully controllable operations, making KOGNIX suitable for enterprise-critical applications.

Automatic Metadata & Entity Tagging

During the ingestion phase, KOGNIX AI Engine employs an LLM-powered pre-processor to detect Named Entities such as dates, names, locations, and project codes. This process generates a structured metadata layer that overlays the unstructured text, enabling more organized and accessible data for downstream retrieval and analysis.

Creating a structured metadata layer during ingestion offers several significant benefits:

- **Enhanced Search and Retrieval**
Metadata enables precise filtering and querying by entities like dates, names, or project codes, improving both relevance and speed of search results.
- **Improved Contextual Understanding**
Linking entities to the text helps the system recognize relationships and context, supporting more accurate reasoning and LLM outputs.
- **Facilitates Data Organization**
Unstructured text is transformed into a more navigable format, making large document collections easier to manage, index, and analyse.
- **Supports Advanced Analytics**
Structured metadata allows for entity-based aggregations, trends analysis, and reporting that would be difficult with raw text alone.
- **Enables Better Compliance and Auditing**
Entity-level metadata provides traceability for sensitive information, helping meet regulatory and audit requirements.
- **Boosts Downstream Workflow Efficiency**
By pre-identifying key entities, subsequent RAG processes, summarization, or reasoning chains can operate faster and more accurately.

In essence, entity extraction and metadata layering turn raw text into actionable, searchable, and analytically valuable information, enhancing both retrieval precision and enterprise usability.

Persistent Memory for Long-Term Dialogue

KOGNIX AI Engine employs a two-tier memory architecture: short-term memory, which uses a buffer to manage immediate conversational context, and long-term memory, which leverages vector embeddings to store information from past sessions. This design enables the agent to remember user preferences and prior conclusions from conversations that occurred weeks earlier.

Using a two-tier memory system provides several important advantages for conversational AI and retrieval workflows:

- **Contextual Continuity**
Short-term memory ensures the agent can maintain coherent responses within the current conversation, preserving context across multiple turns.
- **Historical Awareness**
Long-term memory allows the system to recall past interactions, preferences, and conclusions, enabling more personalized and informed responses over time.
- **Improved User Experience**
Users don't need to repeat prior information or re-establish context, creating smoother and more natural interactions.
- **Enhanced Decision-Making**
Access to historical data and patterns supports better reasoning and more accurate recommendations or answers.
- **Scalability Across Sessions**
Long-term memory ensures knowledge persists beyond individual sessions, making the agent suitable for ongoing projects or long-term enterprise use cases.
- **Efficiency in Knowledge Retrieval**
By distinguishing between immediate context and historical data, the system can optimize retrieval strategies, querying long-term memory only when relevant, reducing computational overhead.

In summary, the two-tier memory system allows to be both context-aware in the moment and historically informed, significantly improving coherence, personalization, and operational intelligence.

Day-to-day KOGNIX AI Engine use cases

To understand how KOGNIX AI Engine fits into everyday work, it is useful to view it through the lens of specific professional roles.

KOGNIX AI Engine is not merely a search interface; it is a context-aware assistant designed to manage the fragmented, unstructured, and often tedious aspects of enterprise information work.

- **Legal & General**
- **Technical Support**
- **Medical Research**
- **Sales & Commercial**
- **Scientific Research**



The Corporate Lawyer: Controlling Contractual Risk and Liability Exposure

- **Operational Challenge**

Senior legal teams routinely review new contract drafts against multiple historical versions to detect subtle changes in liability, indemnification, or limitation clauses. This process is manual, time-intensive, and highly error-prone—particularly when changes are embedded within complex tables or annexes.

- **KOGNIX AI Engine Capability (Document Vision + Semantic Structuring)**

AI Engine ingests all contract versions and reconstructs their visual and logical hierarchy, including clause numbering, tabular structures, and cross-references.

- **Daily Interaction**

The lawyer queries: “Compare indemnity caps in the 2024 draft versus the 2021 executed agreement.”

- **Business and Risk Outcome**

AI Engine surfaces the precise clause and table rows across versions, highlighting a change from a USD 5M to USD 10M liability cap, with direct traceability to source coordinates.

KOGNIX Artificial Intelligence LEGAL & GENERAL INDUSTRY



The Field Engineer: Repairing Equipment with Visual Context

- **Operational Challenge**
Engineers often rely on legacy manuals, scanned drawings, and outdated SOPs when troubleshooting equipment. Errors in interpretation can result in safety incidents, equipment damage, or regulatory non-conformance.
- **KOGNIX AI Engine Capability (Vision-Language Models + On-Prem Deployment)**
Engineers capture images of equipment components, serial plates, or assemblies. AI Engine visually identifies the component and correlates it with validated engineering documentation stored locally.
- **Daily Interaction**
“What is the torque specification for this valve assembly?”
- **Business and Risk Outcome**
The system retrieves the exact diagram and specification from the authoritative manual, with a verifiable citation.



The Medical Researcher: Managing Information Overload

- **Operational Challenge:**
Dozens of new clinical papers are published weekly, making it impractical to read everything while still tracking specific drug interactions.
- **KOGNIX AI Engine Capability (RAPTOR + Hybrid Retrieval):**
New papers are automatically ingested, and KOGNIX AI Engine builds a hierarchical summary of the literature using RAPTOR.
- **Daily Interaction:**
The researcher asks, “Is there a consensus this month on using Drug X in patients with elevated creatinine levels?”

Outcome:

KOGNIX AI Engine analyses high-level summary nodes across all papers, identifies emerging consensus, and points directly to relevant methodology sections—saving time while maintaining scientific rigor

KOGNIX Artificial Intelligence MEDICAL RESEARCH DOMAIN



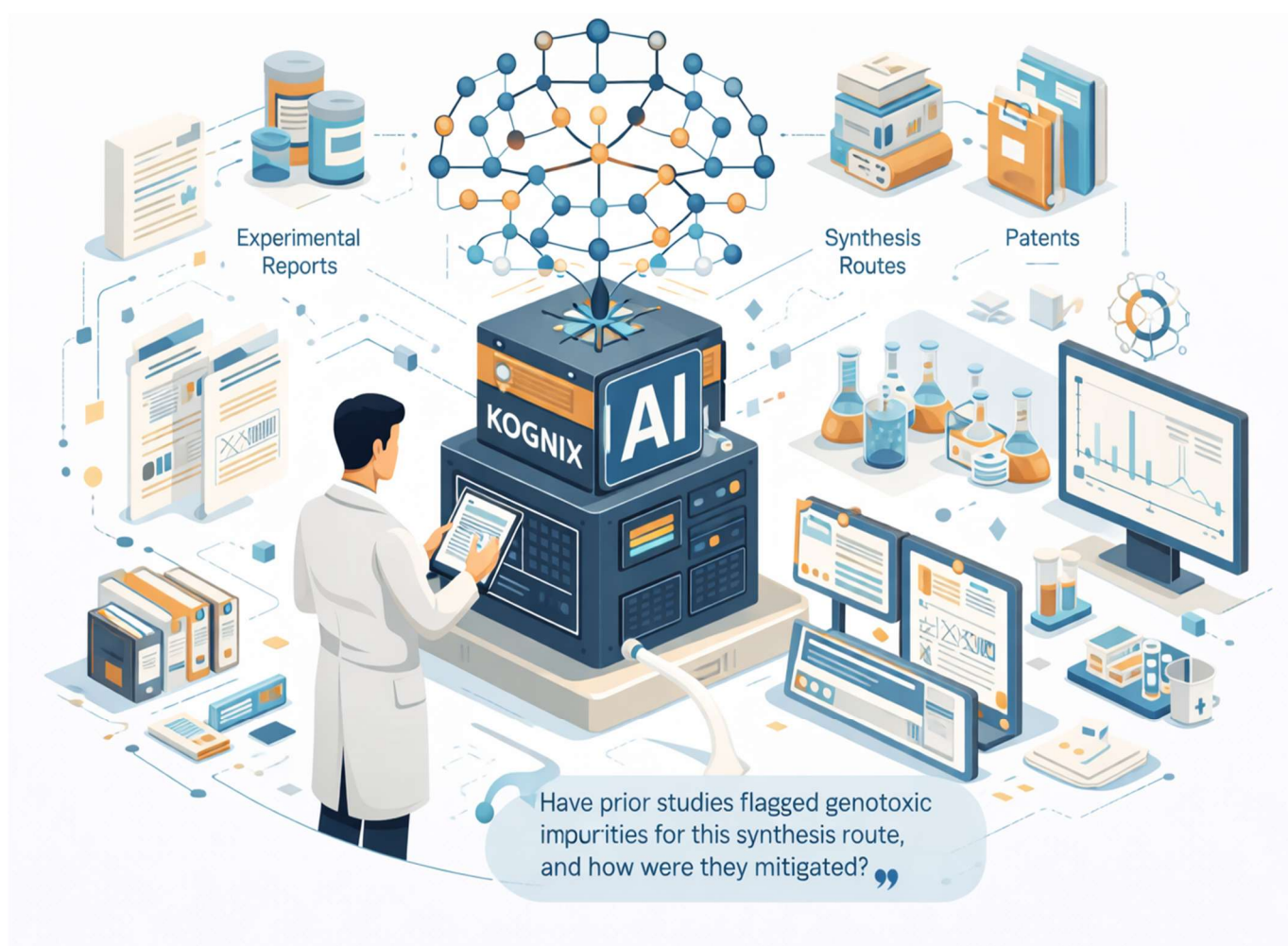
Marketing and Strategy Leader: Operating with Governed Intelligence

- **Operational Challenge**
Leadership teams require timely competitive insights, market intelligence, and internal performance summaries—yet assembling these inputs manually is inefficient and inconsistent.
- **KOGNIX AI Engine Capability (Agentic Workflows + DAG Orchestration)**
Using the low-code workflow editor, teams configure governed agents that combine external intelligence, internal reports, and structured summaries.
- **Daily Interaction**
The strategy workflow executes automatically on a scheduled basis.
- **Business Outcome**
A consolidated, auditable briefing is generated—ready for leadership review.



API Research Scientist: Accelerating Decisions While Meeting Regulatory Expectations

- **Operational Challenge**
API development teams manage years of experimental data across synthesis routes, impurity profiles, batch records, and regulatory submissions. Repeating experiments due to poor knowledge reuse increases cost, delays timelines, and elevates regulatory risk.
- **KOGNIX AI Engine Capability (RAPTOR + Hybrid Retrieval)**
All historical lab reports, development studies, patents, and filings are ingested. AI Engine constructs a hierarchical knowledge model that aligns experimental outcomes, impurity findings, and mitigation strategies.
- **Daily Interaction**
“Have prior development programs identified genotoxic impurity risks for this synthesis route, and how were they controlled?”
- **Business and Regulatory Outcome**
The system identifies relevant historical studies, summarizes mitigation strategies, and links directly to validated lab reports and impurity tables.



Enterprise-First Security and Data Compliance

KOGNIX AI Engine is architected with an enterprise-first, regulation-aware design philosophy, prioritizing data sovereignty, deterministic intelligence, and deep document understanding. Its security posture is built around a single guiding principle: your data always remains under your complete control.

This approach makes KOGNIX AI Engine particularly well-suited for regulated environments, including financial services, legal operations, healthcare, and life sciences.

Below is an overview of the core security and compliance capabilities of KOGNIX AI Engine.

Data Sovereignty and Local Deployment

At the foundation of KOGNIX AI Engine's security model is full support for private, on-premises deployment.

- **Air-Gapped Operation**
KOGNIX AI Engine can be deployed via containerized infrastructure in environments with no external internet connectivity, enabling use in air-gapped or high-security networks.
- **Local Inference and Model Control**
The platform integrates seamlessly with locally hosted inference engines, ensuring that sensitive documents, prompts, and responses are never transmitted to external AI providers or third-party APIs.
- **Infrastructure Ownership**
All underlying components—including retrieval indexes, metadata stores, and document repositories—run entirely within infrastructure owned and managed by your organization. This ensures full control over data storage, access, and lifecycle management.

Data Privacy and Right-to-Erasure Controls

KOGNIX AI Engine includes robust mechanisms to support privacy regulations such as GDPR, CCPA, and similar frameworks.

- **Deterministic Data Deletion**
When a document is deleted, all derived artifacts—including indexed content, embeddings, and metadata—are removed across the system, ensuring complete and verifiable erasure.
- **Conversation and Memory Governance**
Users and administrators can selectively remove conversational memory or historical context. Deleted entries are immediately excluded from retrieval and scheduled for permanent removal, supporting “right to be forgotten” obligations.
- **No External Model Training**
KOGNIX AI Engine does not reuse customer data to train shared or global models. All data remains private, static, and confined to the organization’s-controlled environment.

Enterprise-Grade Security Maintenance and Auditability

- **Ongoing Security Hardening**
KOGNIX AI Engine follows active vulnerability management practices, with regular patches and fixes applied as part of the support and maintenance contract.
- **Grounded, Verifiable Responses**
Every AI-generated output is linked back to its original source content through deterministic citation mapping. This enables auditors, legal reviewers, and compliance teams to independently verify conclusions against authoritative documents.
- This transparency is essential for environments where explainability and traceability are mandatory.

Alignment with Regulatory and Compliance Frameworks

While KOGNIX AI Engine is a deployable enterprise platform rather than a hosted service, its architecture directly supports compliance with common regulatory standards:

Framework	How KOGNIX AI Engine Supports It
GDPR / HIPAA	Fully on-premises deployment and local inference ensure sensitive personal and health data never leaves organizational control.
SOC 2	Provides the technical controls required for audits, including access logging, identity management, and traceable data handling.
Data Residency Laws	Organizations define exactly where data is stored and processed, supporting regional and national data sovereignty requirements.
Regulatory Audits	Deterministic citations and document-level traceability support inspection readiness and defensible AI outputs.

KOGNIX unlocks a new and powerful way of working

KOGNIX AI Engine enables organizations to discover, understand, and act on knowledge with confidence, using enterprise-grade generative AI built for real-world complexity and regulation.

Designed for regulated and knowledge-intensive industries, KOGNIX AI Engine allows teams to connect fragmented information, generate reliable insights, and collaborate securely—across departments, systems, and geographies—without compromising data control or compliance.

By combining deep document intelligence, grounded retrieval, and agentic workflows, KOGNIX AI Engine helps professionals move beyond manual searching and repetitive analysis. It boosts productivity by accelerating research, decision-making, and reporting, while freeing teams to focus on high-value, judgment-driven work.

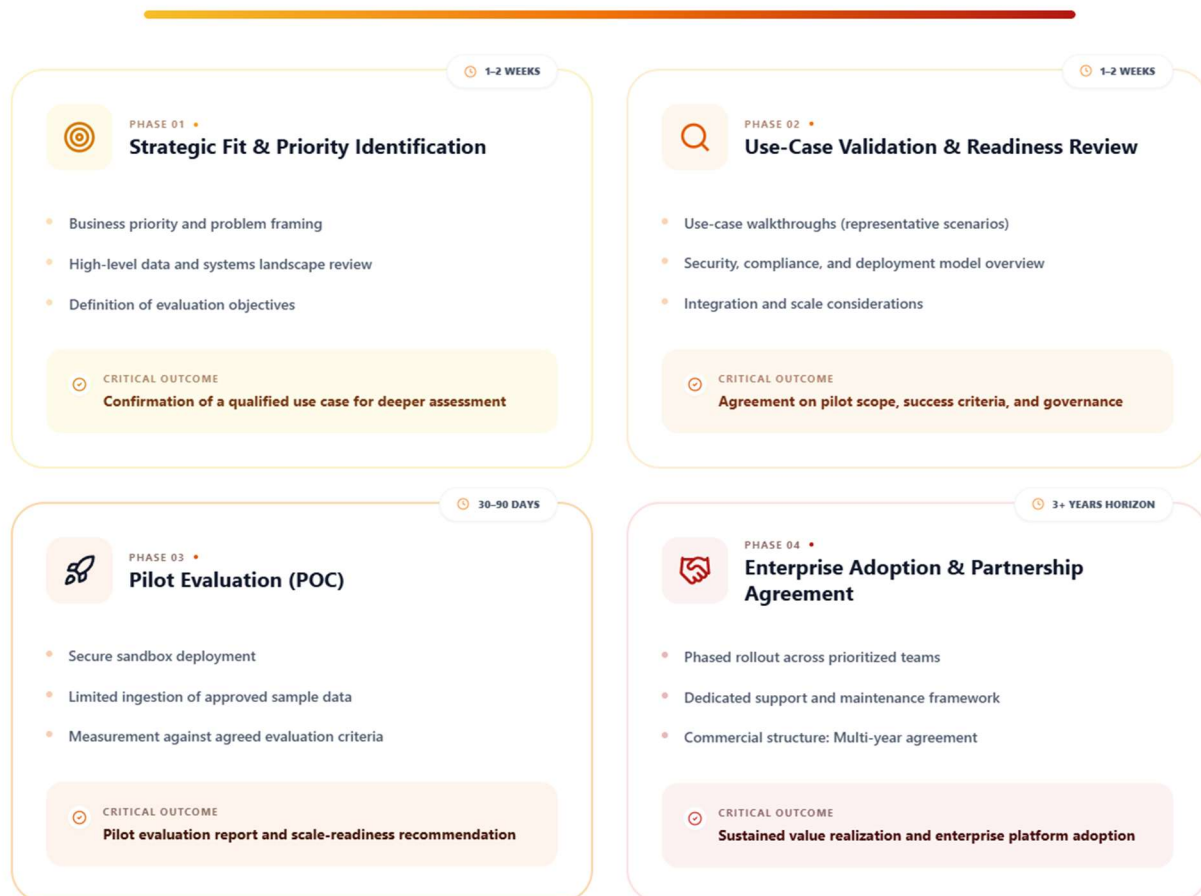
All of this is delivered through a simple, intuitive interface, requiring no code, no disruption to existing workflows, and no dependency on external AI services. KOGNIX AI Engine operates entirely within your enterprise environment, ensuring data sovereignty, auditability, and trust by design.

We are excited to bring KOGNIX AI Engine to enterprises ready to adopt AI with confidence and control—ushering in a new era of responsible, governed intelligence at scale.

To learn how KOGNIX AI Engine can help your organization drive measurable impact while meeting regulatory and security expectations, contact our enterprise team today

Engagement Model

A structured journey from initial strategic discovery to a resilient, long-term enterprise partnership



Enterprise Partnership Horizon

Our model is built on mutual investment. Each stage provides the validation necessary to move confidently toward a long-term, multi-year commercial agreement focused on optimization and scale.

3-Year Cost Comparison: SaaS AI vs KOGNIX AI Suite

Scope Clarification

The USD 500,000 license includes:

- **KOGNIX AI Engine**
- **KOGNIX Analytix**
- **KOGNIX Intelligence**
- Unlimited instances (no cap on deployments)
- 24×7 enterprise support
- All patches, upgrades, and bug fixes
- End-to-end support for:
 - Architecture design
 - Capacity planning
 - Build & deployment
 - Enterprise implementation

This is a single bundled license, not per-module pricing.

Common Assumptions

- **Users:** 1,000 enterprise users
- **Workload:** RAG, Knowledge assistants, Analytics, Multimodal agents
- **Industry:** Regulated (Pharma / Legal / BFSI)
- **Time Horizon:** 3 years
- **Deployment:** Private DataCenter / On-Prem / Air-gapped capable

Option 1: SaaS AI Platforms

(Gemini / OpenAI Enterprise)

Cost Characteristics

- Pure **OpEx**
- Token-metered
- RAG multiplies cost
- Each new workflow increases spend

3-Year Cost Summary (Indicative but Realistic)

Cost Component	Year 1	Year 2	Year 3	3-Year Total
Token Consumption (RAG-heavy usage)	\$45M	\$54M	\$65M	\$164M
Enterprise Support / SLAs	\$0.5M	\$0.6M	\$0.7M	\$1.8M
Compliance & Risk Controls	\$1.2M	\$1.4M	\$1.6M	\$4.2M
Total OpEx	\$46.7M	\$56.0M	\$67.3M	\$170M+

Option 2: KOGNIX AI Suite (Private DataCenter Stack)

ACCURATE COMMERCIAL MODEL

Cost Characteristics

- **CapEx-first, fixed**
- No per-token cost
- No per-user cost
- No per-instance restriction
- No module-wise licensing
- Unlimited internal scale

3-Year Cost Breakdown (UPDATED & FINAL)

Cost Component	Year 1	Year 2	Year 3	3-Year Total
KOGNIX AI Suite License (Perpetual)	\$500,000	—	—	\$500,000
(AI Engine + Analytix + Intelligence, unlimited instances)				
Infrastructure (GPU, Storage, DC)	\$300k–\$500k	—	—	\$300k–\$500k
Architecture, Build, Capacity Planning	Included	Included	Included	\$0
Enterprise Support, Patches, Upgrades	Included	Included	Included	\$0
Total 3-Year Cost				\$800k–\$1.0M

Side-by-Side: What Actually Changes for the Enterprise

Dimension	SaaS AI Platforms	KOGNIX AI Suite
Licensing Model	Per-token, per-usage	Perpetual, bundled
Modules Included	Charged separately	AI + Analytics + Intelligence
Instance Limits	Yes	Unlimited
Token Billing	Yes	None
User Limits	Often capped	Unlimited
RAG Cost Impact	Extremely high	Zero marginal cost
Architecture Support	Minimal	Included end-to-end
3-Year Spend	\$170M+	\$0.8M–\$1.0M
Cost Predictability	Low	Absolute

Break-Even Reality

KOGNIX typically breaks even against SaaS AI within the first month of production usage. Every additional user, workflow, or instance after that is effectively free.

“SaaS AI platforms monetize every question.

KOGNIX monetizes the platform once and lets the enterprise scale without penalty.”

Why This Naturally Supports a 3-Year (and Longer) Contract

- One-time license already covers multi-year use
- Unlimited expansion without renegotiation
- No pricing surprises
- Infrastructure becomes an owned strategic asset
- Switching risk is lower than SaaS lock-in

Team KOGNIX
VAI LABs
sales@vailabs.in
www.vailabs.in